

4.3 Interval Estimation

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4.3.1 Revisiting Sampling Distributions

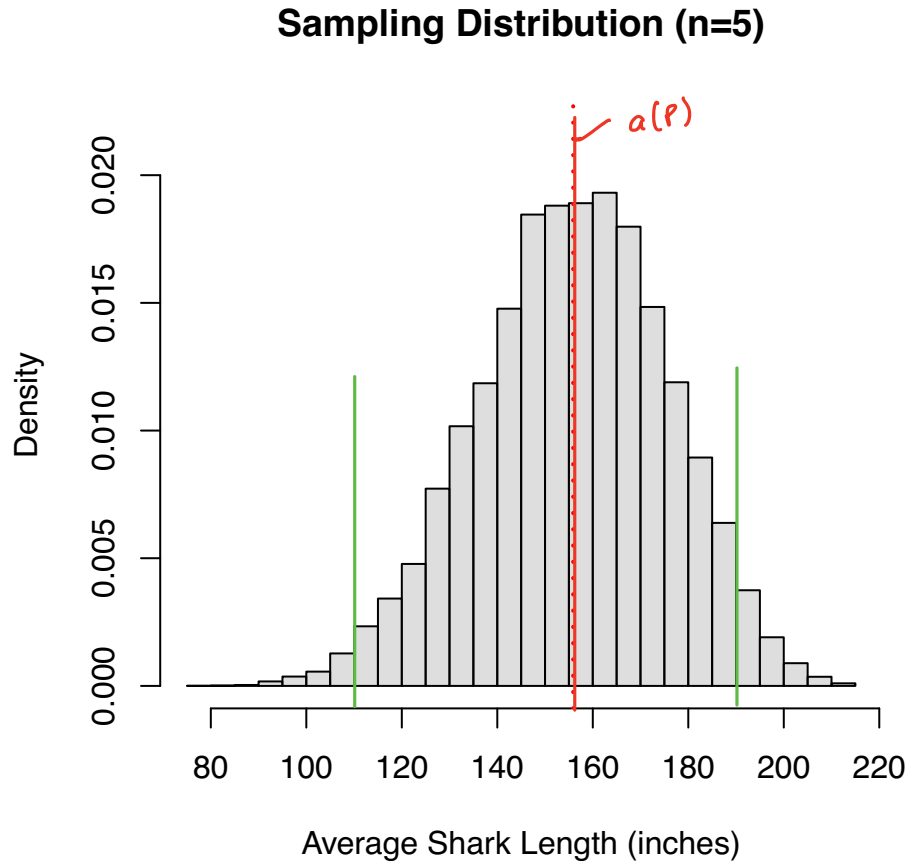
Overview

- As an introduction to Interval Estimation and Coverage Probabilities we revisit Sampling Distributions
-

- Recall that when we look at $a(\mathcal{S})$ for all possible samples $\mathcal{S}$ of some size n from a population $\mathcal{P}$ and that the values of $a(\mathcal{S})$ have a distribution.
 - We called this the **sampling distribution** of $\tilde{a}(\mathcal{S})$.

* For example, we looked at the Australian shark encounter population ($N = 28$), and we considered all possible samples of size $n = 5$.

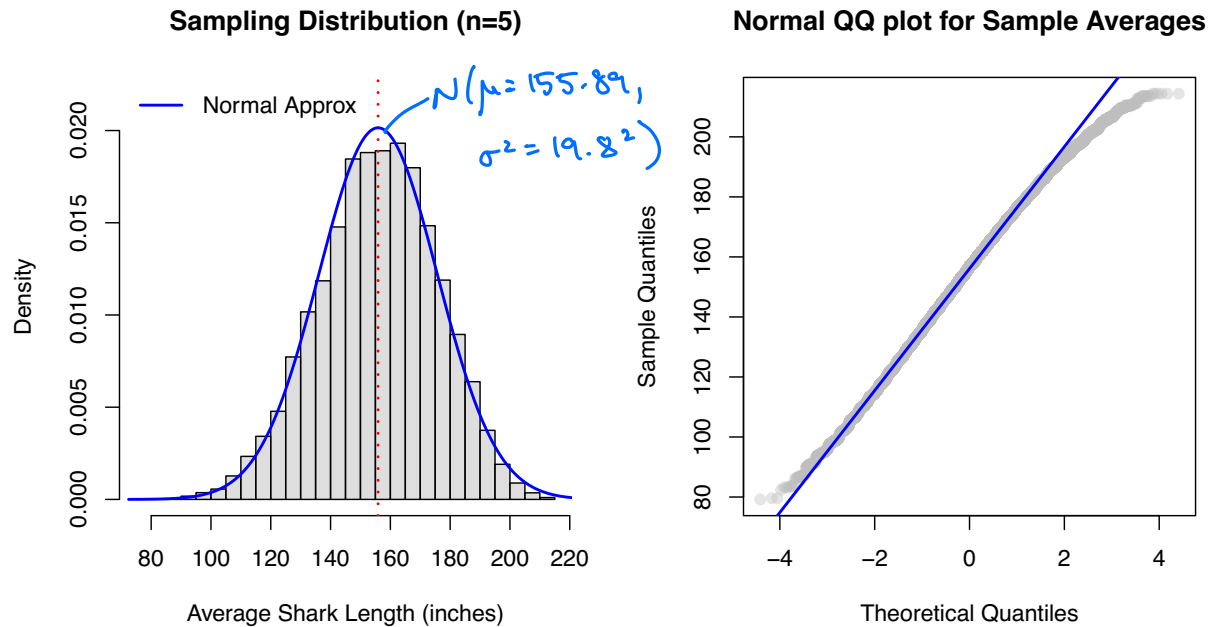
- We calculated $a(\mathcal{S}) = \bar{y}$, the average shark length, for each sample and then constructed a histogram.



- The histogram is fairly symmetric and centered about the population average $a(\mathcal{P}) = 155.89$
 - The 2.5th and 97.5th quantiles of this distribution were 115.4 and 192.8
 - Thus we know that the interval (115.4, 192.8) contains the 95% of the possible sample averages.
- Quantifying the spread in this way is very useful – but in order to do so exactly we require having observed **all possible samples**.
 - Alternatively, we could instead consider a large number of possible samples to approximate the sampling distribution.
 - * Or, we could approximate the sampling distribution via the normal distribution to simplify calculations.
 - * Or, we could approximate the sampling distribution via resampling techniques such as the **bootstrap** (see Section 4.4).



- The normal distribution that best approximates the sampling distribution is the one with mean and standard deviation equal to the mean, $\text{mean}(\text{avesSamp}) = 155.89$ and standard deviation, $\text{sd}(\text{avesSamp}) = 19.8$, from all possible $a(S)$ values.
 - The plots below evaluate the adequacy of this approximation.



- The plots suggest that the normal approximation is pretty good, except for in the tails.
 - This is apparent in both in the histogram and the QQ-plot.
 - * The closeness of this approximation is unsurprising – recall the **central limit theorem (CLT)** and its implications for distributions of averages

* The normal approximation provides a model for the sampling distribution and can be used as a basis to construct confidence intervals for population averages. (and an attribute that can be expressed as a total).

* BUT many attributes will have a sampling distribution that is not approximately normal, so we're going to need another method too.

Summary

- The sampling distribution of some attributes can be approximated by a normal distribution.

4.3.2 Random vs. Observed Intervals

Overview

- We see how to construct a confidence interval, and we emphasize the distinction between random and observed intervals.

- Suppose the attribute of interest is the population average $a(\mathcal{P}) = \mu = \frac{1}{N} \sum_{u \in \mathcal{P}} y_u = \bar{y}$

- Recall (from STAT 231) that the estimator $\tilde{a}(\mathcal{S}) = \tilde{\mu} = \bar{Y}$ (a random variable) has the following properties:

$$E[\bar{Y}] = \mu \quad \text{and} \quad \text{Var}[\bar{Y}] = \frac{\sigma^2}{n}$$

where

$$\sigma^2 = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \mu)^2$$

is the population variance.

- If the normality assumption holds (this may be appropriate due to the CLT) then the estimator $\tilde{a}(\mathcal{P}) = \bar{Y}$ has the following distribution:

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- In the case of the Australian shark data

```
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)
popSharksAustralia <- popSharks[sharks$Australia == 1]

samples <- combn(popSharksAustralia, 5)
M <- ncol(samples)

avePop <- mean(sharks[popSharksAustralia, "Length"])
avesSamp <- apply(samples, MARGIN = 2,
  FUN = function(s){mean(sharks[s, "Length"])})

sdN = function(x){sqrt(var(x)*(length(x)-1)/length(x))}
```

- Some attributes were

- $N = 28$ and $n = 5$
- $\mu = a(\mathcal{P}) = 155.89$ is the population average
- $\sigma = 47.97$ is the population standard deviation
- $SD[\bar{Y}] = \frac{\sigma}{\sqrt{n}} = 19.8$.
- Standardizing this random variable yields

$$Z = \frac{\bar{Y} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1)$$

- **Note:** Strictly speaking we are dealing with a finite population and so our variance term should include the finite population correction:

$$Var[\bar{Y}] = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) \quad \text{FPC}$$

Exercise in
S3.2.5a

(Recall this was derived in the context of the variance of the Horvitz-Thompson estimator under SRSWOR)

- However, when $N \gg n$ then $\frac{N-n}{N-1} \rightarrow 1$ in which case this term can be ignored
- This is why most texts omit the correction factor, giving rise to the more familiar formulation $Var[\bar{Y}] = \frac{\sigma^2}{n}$
- For completeness we will henceforth include the finite population correction, in which case

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)\right)$$

and

$$Z = \frac{\bar{Y} - \mu}{\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} \sim N(0, 1)$$

- A histogram and corresponding $N(0, 1)$ curve for this standardized quantity is visualized below:

```
N <- 28
n <- 5
fpc <- (N-n)/(n-1)
Z = (avesSamp - mean(avesSamp)) / sdN(avesSamp)

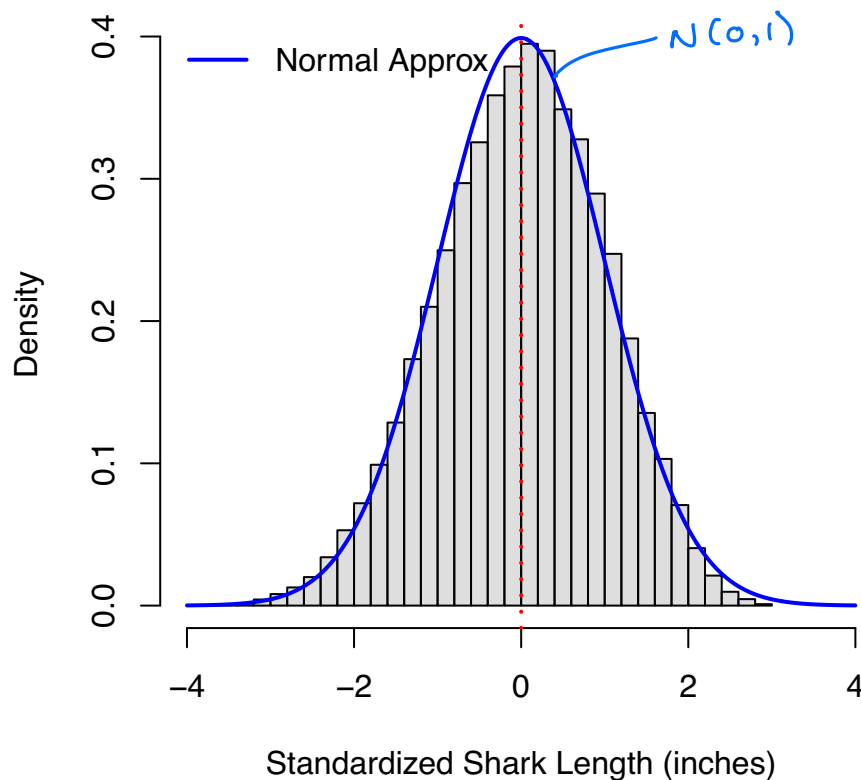
hist(Z, col=adjustcolor("grey", alpha = 0.5),
     freq = FALSE,
     main="Sampling Distribution (n = 5)",
     xlab="Standardized Shark Length (inches)",
     breaks=25,
     xlim = c(-4,4))

### Mark the population attribute in red
abline(v=0, col="red", lty=3, lwd=2)

### Add a Gaussian density
x = seq(-4, 4, length.out=1000)
lines(x, dnorm(x), col = "blue", lwd = 2)

legend("topleft", legend = c("Normal Approx"), lwd = 2,
      col = "blue", lty = 1, bty = "n")
```

Sampling Distribution (n = 5)



Random Intervals

- Using the standardized random variable and specified $p \in (0, 1)$ we can find a constant $c > 0$ such that

$$1-p = \Pr \left(-c < \frac{\bar{Y} - \mu}{\frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} < c \right)$$

$$= \Pr \left(\bar{Y} - c \times \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} < \mu < \bar{Y} + c \times \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right)$$

- Rearranging this statement yields a **random interval** which contains μ with probability $1 - p$.

$$\left[\bar{Y} - c \times \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{Y} + c \times \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

- This interval is considered **random** because it's never actually observed – it is defined in terms of random variables

↑ specifically $\bar{Y} = \tilde{a}(s)$

- However, observed intervals calculated by substituting $\bar{Y}$ with $\bar{y}$ are guaranteed to contain μ , $100(1 - p)\%$ of the time.
 - $1 - p$ is therefore called the **coverage probability**.
 - * μ is contained in (or *covered* by) such an interval $100(1 - p)\%$ of the time.
 - All these intervals have the same width, they just have different (random) centres.

A Note on determining c :

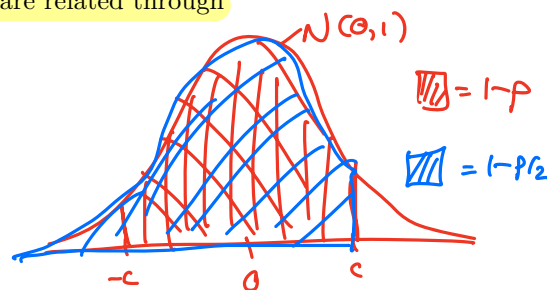
- The normal distribution is symmetric about its mean μ , so p and c are related through

$$1 - p = \Pr(-c \leq Z \leq c)$$

or, equivalently,

$$1 - p/2 = \Pr(Z \leq c)$$

where $Z \sim N(0, 1)$ is a standard normal random variable.



- Therefore, given any $p \in (0, 1)$ the value of c can be determined from the quantile function of a standard normal random variable:

$$c = Q_Z \left(1 - \frac{p}{2} \right)$$

which in R is calculated as `qnorm(1 - p/2)`.

- e.g., $c \approx 1.96$ when $1 - p = 0.95$ for a standard normal random variable.

this is one we can actually calculate given a sample of data

Observed Intervals

- In practice, we will have only one sample
 - And thus a single numerical average $\bar{y}$
 - And thus a single instance of these randomly generated intervals:

$$\left[\bar{Y} - c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{Y} + c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

– In particular, we observe

$$\left[\bar{y} - c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{y} + c \times \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

known known

- Such observed intervals are referred to as **confidence intervals** and you must take care to distinguish them conceptually from a random interval
 - * In the context of random intervals, probabilistic statements are sensible.
 - However, an observed confidence interval is not random – it either contains μ or it doesn't.
- Probability your interval contains μ is 0 or 1.
- Probability statements are in reference to the method used to generate the intervals, NOT to the particular interval we have observed.

- * If the normality assumption holds up, $100(1-p)\%$ of such intervals will contain μ ,
 - We thus have some **confidence** that our particular observed interval will contain μ as well, but unfortunately we'll never know if it does
- * The larger $1-p$, the more confident we are that the interval will contain μ .

Example: Coverage Probability Estimation via Simulation

- Given that $\mu = a(\mathcal{P}) = 155.89$ and $SD(\bar{Y}) = \text{sdN}(\text{avesSamp}) = 19.8$.
 - We randomly select 100 samples of size $n = 5$
 - * For each sample we calculate the 95% confidence interval (i.e., $(1-p) = 0.95$ and $c \approx 1.96$)
 - We should hope to find that approximately $100(1-p)\% = 95\%$ of the intervals contain (i.e., cover) $\mu = 155.89$.


```

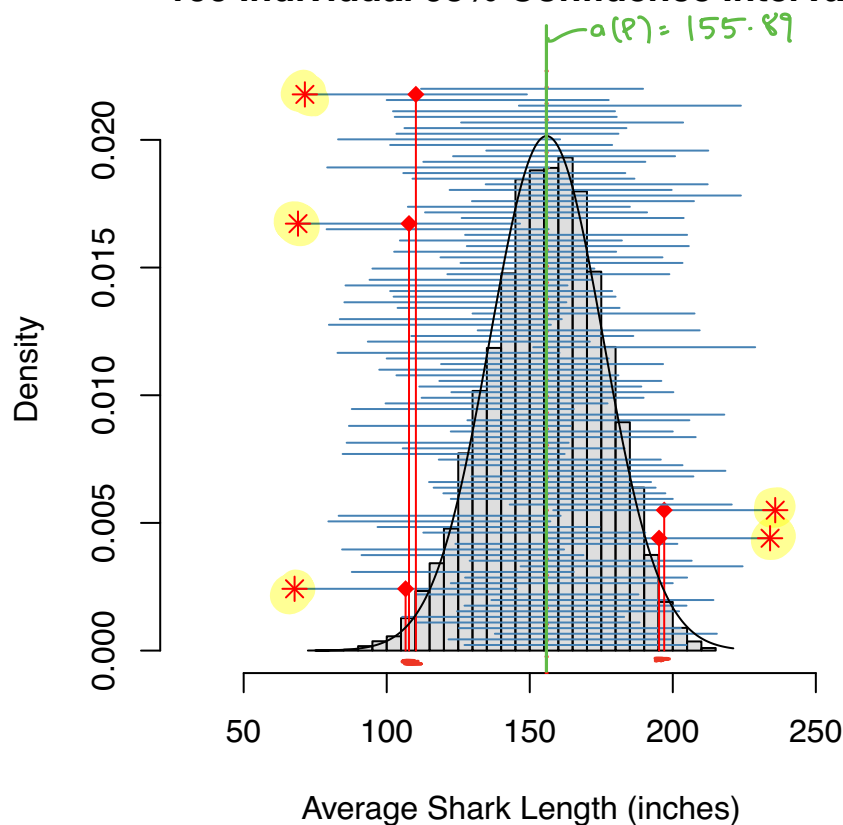
## R code to simulate the GAUSSIAN approximation
ylim <- c(0, 0.022)

### Add a Gaussian density
tmpAve <- mean(avesSamp)
tmpSD <- sdN(avesSamp)
confidence <- 0.95
p <- 1 - confidence
numIntervals <- 100
cValue <- qnorm(1 - p/2) # or qnorm((confidence + 1)/2)
set.seed(341) # comment out this line to get different samples every time you run the code.
ybarSampled <- sample(avesSamp, numIntervals)
heights <- seq(diff(ylim)/numIntervals, max(ylim), length.out = numIntervals)
xlim <- extendrange(avesSamp + cValue * c(-tmpSD, tmpSD))
hist(avesSamp, col=adjustcolor("grey", alpha = 0.5), freq = FALSE,
     main=paste0(numIntervals, " Individual ", round(100 * confidence), "% Confidence Intervals"),
     xlab="Average Shark Length (inches)",
     ylim=ylim, xlim = xlim,
     breaks=25
    )

### Mark the population attribute in red
abline(v=avePop, col="red", lty=3, lwd=2)
numIntervalsMissed <- 0
for(i in 1:numIntervals) {
  lines(ybarSampled[i] + cValue * c(-tmpSD, tmpSD), rep(heights[i],2),
        col = "steelblue")
  if (tmpAve > ybarSampled[i] + cValue*tmpSD) {
    points(ybarSampled[i] - cValue*tmpSD, heights[i], pch=8, cex=1.2, col="red")
    points(ybarSampled[i], heights[i], pch=18, cex=1.2, col="red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else if (tmpAve < ybarSampled[i] - cValue*tmpSD) {
    points(ybarSampled[i] + cValue*tmpSD, heights[i], pch=8, cex=1.2, col="red")
    points(ybarSampled[i], heights[i], pch=18, cex=1.2, col="red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else numIntervalsMissed <- numIntervalsMissed + 1
}
tmpX <- extendrange(avesSamp)
tmpX <- seq(tmpX[1], tmpX[2], length.out = 200)
lines(tmpX, dnorm(tmpX, mean = tmpAve, sd = tmpSD))

```

100 Individual 95% Confidence Intervals



- 95 of these 100 intervals cover the value μ (which is indicated by the vertical dashed red line)
 - Those which *do not* cover μ are marked with an asterisk and the corresponding sample means are indicated by diamonds and vertical solid red lines.
 - What do we notice?

↳ The problematic samples have averages coming from the tails of the sampling distribution

Summary

- We saw that a confidence interval is a realization of a random interval.
- Probability statements, such as “95% of the random intervals contain the true value”, are in reference to the method used to generate the intervals, NOT to the particular interval we have observed.
- The confidence intervals we calculated assumed $SD[\bar{Y}]$ was known which is an unrealistic assumption.

4.3.3 Student t Based Intervals

Overview

- We discuss how to construct a confidence interval using a t -distribution as an approximation to the sampling distribution.

Standard Error vs. Standard Deviation

$$\sigma \rightarrow \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

- In the previous section the confidence intervals we calculated assumed $SD[\bar{Y}]$ was known
 - This is an unrealistic assumption.
 - Only very rarely would we have this value.

* However, for many sample attributes $a(\mathcal{S})$ (e.g., Horvitz-Thompson estimators), we can estimate the standard deviation $SD[\tilde{a}(\mathcal{S})]$ of the **sampling distribution** of $\tilde{a}(\mathcal{S})$.

- The **standard error** is an estimate of the standard deviation of the corresponding estimator:

$$SE[\tilde{a}(\mathcal{S})] = \hat{SD}[\tilde{a}(\mathcal{S})]$$

- We could use $\frac{a(\mathcal{S}) - a(\mathcal{P})}{SE[\tilde{a}(\mathcal{S})]}$ instead of $\frac{a(\mathcal{S}) - a(\mathcal{P})}{SD[\tilde{a}(\mathcal{S})]}$ as pivotal quantities

- Note that using the estimate SE in place of SD will increase the variability of the random intervals.
- The corresponding estimator has much more variability – since we have to estimate $\widehat{SD}$ now as well.

- **Example:** Let us illustrate this phenomenon using the Australia shark lengths data
 - Because we've taken all possible samples, we know $SD[\bar{a}(S)]$ but let's pretend like we don't and estimate it.
 - Thus let's calculate both $\hat{\sigma}$ and $\hat{\sigma}_n$

```
### Loading the shark data
sharks <- read.csv("../Data/sharks.csv")
popSharks <- rownames(sharks)
popSharksAustralia <- popSharks[sharks$Australia == 1]

samples <- combn(popSharksAustralia, 5)
M <- ncol(samples)

avePop <- mean(sharks[popSharksAustralia, "Length"])
avesSamp <- apply(samples, MARGIN = 2,
                  FUN = function(s){mean(sharks[s, "Length"])}))

sdN = function(x){sqrt(var(x)*(length(x)-1)/length(x))}
# a(S) a(P) SD[a(S)]
Z = (avesSamp - mean(avesSamp))/sdN(avesSamp)

sdsSamp <- apply(samples, MARGIN = 2,
                  FUN = function(s){sdN(sharks[s, "Length"])}))

n = 5
N = 28
se = sdsSamp/sqrt(n)*sqrt((N-n)/(N-1))
# sigma-hat/sqrt(n) * sqrt((N-n)/(N-1))
t = (avesSamp - mean(avesSamp))/se
# a(S) a(P) SE[a(S)]

## So the two histograms have the same bins
delta = 0.2
brk = seq(min(t)-delta, max(t)+delta, delta)

par(mfrow=c(1,2))
hist(Z, col=adjustcolor("grey", alpha = 0.5),
     freq = FALSE,
     main="Standardized averages \n with known standard deviation",
     xlab="Standardized Shark Length (inches)",
     breaks=brk, xlim=c(-4,4), ylim=c(0,0.4) )

### Mark the population attribute in red
abline(v=0, col="red", lty=3, lwd=2)

### Add a Gaussian density
x = seq(-4, 4, length.out=1000)
lines( x, dnorm(x) , lwd = 2, col = "blue")

hist(t, col=adjustcolor("grey", alpha = 0.5),
     freq = FALSE,
     main="Standardized averages \n with estimated standard deviation",
```

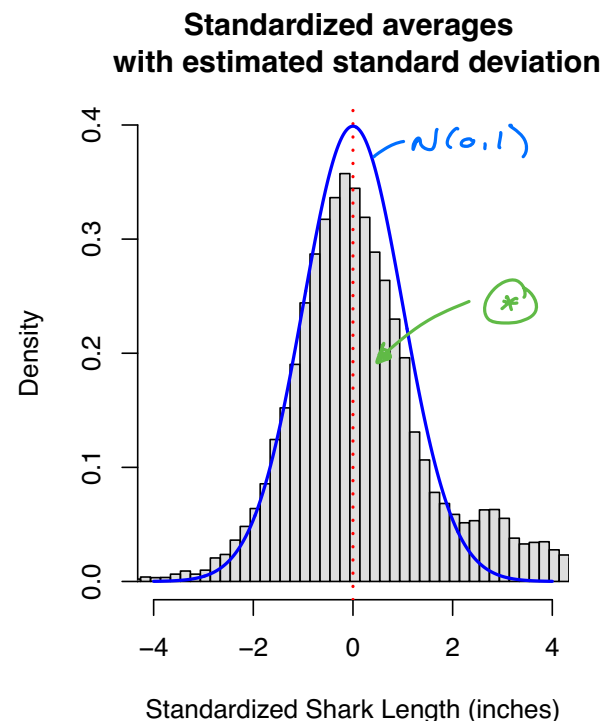
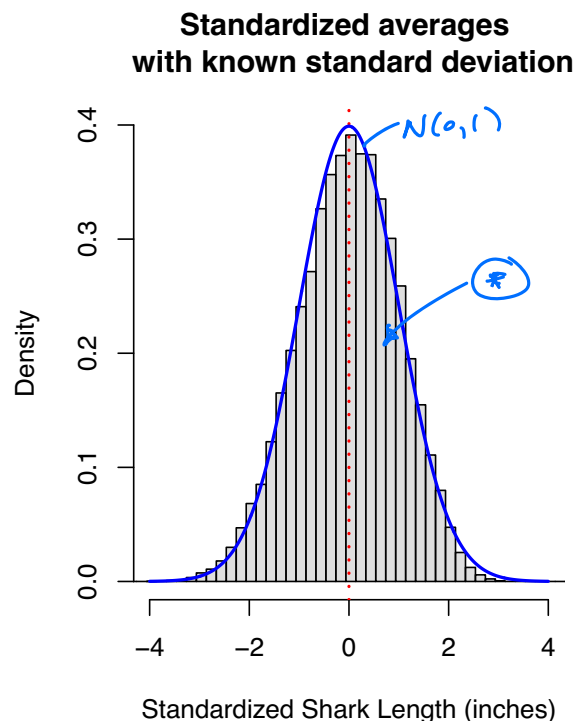
```

xlab="Standardized Shark Length (inches)",
breaks=brk, xlim=c(-4,4), ylim=c(0,0.4) )

### Mark the population attribute in red
abline(v=0, col="red", lty=3, lwd=2)

### Add a Gaussian density
x = seq(-4, 4, length.out=1000)
lines( x, dnorm(x) , lwd = 2, col = "blue")

```



- The blue lines correspond to the $N(0,1)$ density curve
 - the normal distribution does not seem to be a good approximation when we have to estimate the standard deviation.

- If the data are normally distributed we have the following distributional result

$$T = \frac{\bar{Y} - \mu}{\frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}} \sim t_{(n-1)}$$

- This statistic is called a **pivotal quantity** because
 - it is a function of the sample data ($Y_u, u \in \mathcal{S}$) and unknown parameter (μ) and its **sampling distribution is completely known**.

- Let's try the t_{n-1} distribution as an approximation instead of the $N(0, 1)$:

```
n = 5
N = 28
se = sdsSamp/sqrt(n)*sqrt((N-n)/(N-1))
t = (avesSamp - mean(avesSamp))/se

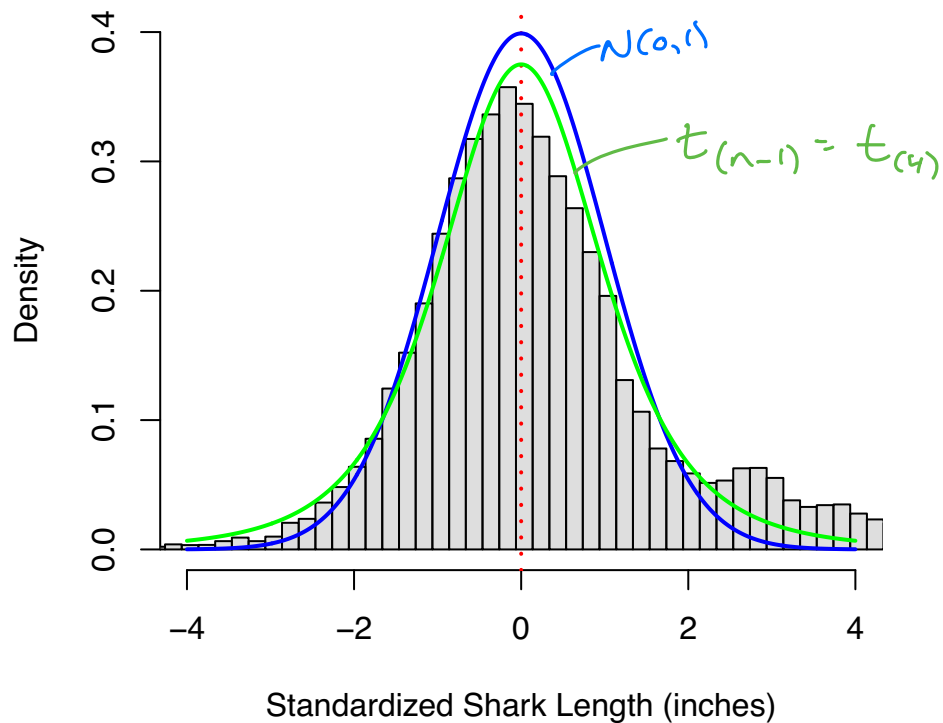
## So the two histograms have the same bins
delta = 0.2
brk = seq(min(t)-delta, max(t)+delta, delta)

hist(t, col=adjustcolor("grey", alpha = 0.5),
     freq = FALSE, ylim = c(0,.4),
     main="Standardized averages with \n estimated standard deviation",
     xlab="Standardized Shark Length (inches)",
     breaks=brk, xlim=c(-4,4) )

### Mark the population attribute in red
abline(v=0, col="red", lty=3, lwd=2)

### Add a Gaussian density
x = seq(-4, 4, length.out=1000)
lines( x, dnorm(x) , col = "blue", lwd=2 )
lines( x, dt(x, n-1), col= "green", lwd=2)
```

Standardized averages with estimated standard deviation



- The green and blue curves represent the t -distribution and normal distribution density curves, respectively
 - The t -distribution with $n - 1 = 4$ is a better approximation, but still not great.

- Now if we suppose that
$$\frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widetilde{SD}[\tilde{a}(\mathcal{S})]} \sim t_{(n-1)}$$

- Then we choose a $p \in (0, 1)$ and a corresponding $c > 0$ with

$$\begin{aligned} 1 - p &= Pr \left(-c \leq \frac{\tilde{a}(\mathcal{S}) - a(\mathcal{P})}{\widetilde{SD}[\tilde{a}(\mathcal{S})]} \leq c \right) \\ &= Pr \left(\tilde{a}(\mathcal{S}) - c \times \widetilde{SD}[\tilde{a}(\mathcal{S})] \leq \mu \leq \tilde{a}(\mathcal{S}) + c \times \widetilde{SD}[\tilde{a}(\mathcal{S})] \right) \end{aligned}$$

we're pivoting
↙
↖ $a(\mathcal{P})$

- This random interval has both a random center and a random length.

- The value of c is determined using the t distribution with $n - 1$ degrees of freedom.

- In R use `qt(1-p/2, df = n-1)` to get the value of c
 - You can check that $c \approx 2.78$ when $1 - p = 0.95$ for a t_4 random variable.

$$c = Q_{\tau}(1 - p/2)$$

- In the special case that $a(\mathcal{P}) = \bar{Y}$ is the population average then the standard deviation is

$$SD[\tilde{a}(\mathcal{S})] = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N - n}{N - 1}} \quad \leftarrow \text{true unknown value}$$

and the standard deviation estimator is

$$\widetilde{SD}[\tilde{a}(\mathcal{S})] = \frac{\tilde{\sigma}}{\sqrt{n}} \times \sqrt{\frac{N - n}{N - 1}} \quad \leftarrow \text{random variable (estimator)}$$

and the standard deviation estimate (the **standard error**) is

$$\widehat{SD}[\tilde{a}(\mathcal{S})] = \frac{\hat{\sigma}}{\sqrt{n}} \times \sqrt{\frac{N - n}{N - 1}} \quad \leftarrow \text{estimated (observed) value}$$

↑
SE

- Thus the random interval in this case is

$$\left[\bar{Y} - c \times \frac{\tilde{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{Y} + c \times \frac{\tilde{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right].$$

- And the corresponding observed confidence interval is

$$\left[\bar{y} - c \times \frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{y} + c \times \frac{\hat{\sigma}}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right].$$

- **Note:** In the formulae above σ denotes the population standard deviation, which may be estimated in a sample by

$$\hat{\sigma} = \sqrt{\frac{\sum_{u \in \mathcal{S}} (y_u - \bar{y})^2}{n}}$$

Example: Coverage Calculation via Simulation

- We randomly select 100 samples of size $n = 5$, and
- For each sample we calculate the 95% confidence interval (defined above)
 - using $c \approx 2.78$ from the t_4 distribution
- We should hope to find that approximately 95% of the intervals contain (or cover) $\mu = 155.89$.

```
## R code to simulate the t-STUDENT approximation
ylim <- c(0, 0.022)
n <- 5
N <- 28

### Add a Gaussian density
tmpAve <- mean(avesSamp)
tmpSD <- sdN(avesSamp)
confidence <- 0.95
numIntervals <- 100
cValue <- qt((confidence + 1)/2, df = n-1)
set.seed(341)
tmpSamp <- sample(1:length(avesSamp), numIntervals)
ybarSampled <- avesSamp[tmpSamp]
sdSampled <- sdsSamp[tmpSamp]
maxSD <- max(sdSampled)/sqrt(n)

heights <- seq(diff(ylim)/numIntervals, max(ylim), length.out = numIntervals)
xlim <- extendrange(c(ybarSampled - cValue * sqrt((N-n)/(N-1)) * sdSampled/sqrt(n),
                     ybarSampled + cValue * sqrt((N-n)/(N-1)) * sdSampled/sqrt(n)))
hist(avesSamp, col=adjustcolor("grey", alpha = 0.5), freq = FALSE,
     main=paste0(numIntervals, " Individual ", round(100 * confidence), "% Confidence Intervals"),
     xlab="Average shark length (inches)",
     ylim=ylim, xlim = xlim,
```



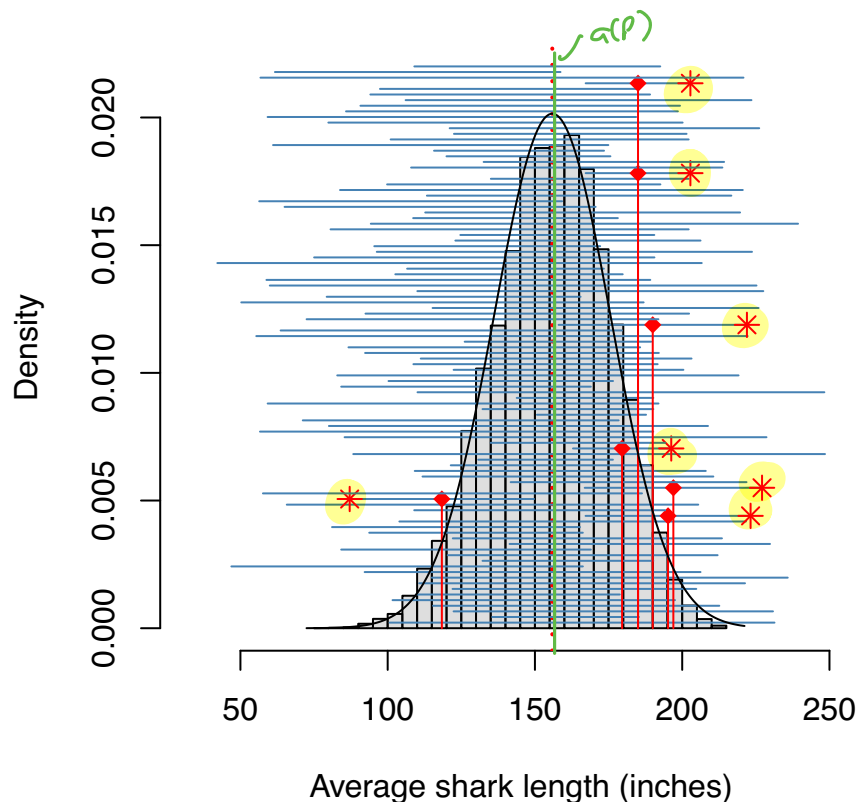
```

breaks=25)
### Mark the population attribute in red
abline(v=avePop, col="red", lty=3, lwd=2)
numIntervalsMissed <- 0
for(i in 1:numIntervals) {
  tmpSampSD <- sdSampled[i]/sqrt(n)*sqrt((N-n)/(N-1))

  lines(ybarSampled[i] + cValue * c(-tmpSampSD, tmpSampSD), rep(heights[i],2),
        col = "steelblue")
  if (tmpAve > ybarSampled[i] + cValue*tmpSampSD) {
    points(ybarSampled[i] - cValue*tmpSampSD, heights[i], pch=8, cex=1.2, col="red")
    points(ybarSampled[i], heights[i], pch=18, cex=1.2, col="red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else if (tmpAve < ybarSampled[i] - cValue*tmpSampSD) {
    points(ybarSampled[i] + cValue*tmpSampSD, heights[i], pch=8, cex=1.2, col="red")
    points(ybarSampled[i], heights[i], pch=18, cex=1.2, col="red")
    lines(rep(ybarSampled[i], 2), c(0, heights[i]), col = "red")
  } else numIntervalsMissed <- numIntervalsMissed + 1
}
tmpX <- extendrange(avesSamp)
tmpX <- seq(tmpX[1], tmpX[2], length.out = 200)
lines(tmpX, dnorm(tmpX, mean = tmpAve, sd = tmpSD))

```

100 Individual 95% Confidence Intervals



- 93 of these 100 intervals cover the value μ (which is indicated by the vertical dashed red line)
 - Those which *do not* cover μ are marked with an asterisk and the corresponding sample means are indicated by diamonds and vertical solid red lines.
 - What do we notice?
Problematic intervals come from samples with means far from $a(P)$.
- **Note** that we can find the exact coverage probability by calculating all possible confidence intervals.

A Note on Pivotal Quantities:

- Pivotal quantities are the basis for constructing random intervals.
- Many random intervals are constructed via pivotal quantities such as

$$\frac{\tilde{a}(S) - a(P)}{\tilde{SD}[\tilde{a}(S)]}$$

- The term *pivot* comes from the fact that with this quantity (which is a function of both S and P) we are able to pivot and isolate for $a(P)$
 - This is the general prescription for constructing random intervals.
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- This particular pivotal is a *t*-like pivotal, but it's not the *only* form of pivotal quantity.
 - One common pivotal quantity used for scale attributes $s(\cdot)$, are of the form

$$\frac{\tilde{s}(S)}{s(P)}$$

- e.g., this gives rise to χ^2 -based intervals for population variances.

So what have we learned?

- Random intervals for population attributes can be constructed by picking an appropriate pivotal quantity and pivoting.
- The corresponding CI is obtained by substituting sample estimates for estimators.
- However, all of this assumes we know the distribution of the pivotal quantity.....

Summary

- We reviewed the notion of a pivotal quantity.
- We used a t-distribution as an approximation to the sampling distribution of the pivotal quantity to construct a confidence interval.